

Week 4 Deliverable: Statistical Model Selection and Hypothesis Development

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Overview

Building on the Week 3 EDA, this document presents our model selection rationale, feature choices, and formal hypotheses for the Star Power Index project, which examines whether cast popularity independently predicts global box office revenue beyond what production budget alone explains.

Statistical Technique Justification

We selected three complementary methods based on our data structure and research questions.

Table 1

Statistical Techniques Selected

Technique	Applied To	Justification
Multiple Linear Regression (OLS)	log(Revenue) as outcome	Tests whether log(Cast Popularity) independently predicts revenue after controlling for budget and vote average. Oh et al. (2017) validated this approach for social media-to-revenue analysis.
One-Way ANOVA	Revenue across genre groups	Tests whether mean revenue differs significantly across the eight primary genre categories observed in

		the Week 3 box plots. Shahid and Islam (2023) showed genre moderates cast-feature effects.
Logistic Regression	ROI > 1.0 (binary)	Classifies whether a film earns more than double its budget. Bogaert et al. (2021) found that volume-based social signals improve profitability classification over budget-only models.

Feature Selection

All continuous predictors are log-transformed to address the right skew confirmed in Week 3. The final MLR feature set is below.

Table 2

Selected Features for Multiple Linear Regression

Feature	Expected Sign	Reason for Inclusion
log(Cast Popularity Score)	Positive	Core experimental variable (SPI). Correlated with log(revenue) at $r = 0.48$ in EDA, independent of budget ($r = 0.35$ between the two).
log(Budget)	Positive	Strongest predictor ($r = 0.72$). Must be controlled to isolate the cast popularity effect.
Vote Average	Positive	Controls for audience quality signal ($r = 0.21$). Prevents quality effects from being misattributed to cast popularity.
Release Season (dummies)	Varies	Accounts for seasonal revenue variation. Spring is the reference category.

Hypothesis

One hypothesis set per statistical test:

H1 (MLR): H0 states that the coefficient on $\log(\text{Cast Popularity})$ equals zero after controlling for budget, vote average, and season. H1 states that the coefficient is significantly different from zero and positive. Test: two-tailed t-test, $\alpha = 0.05$.

H2 (ANOVA): H0 states that mean $\log(\text{Revenue})$ is equal across all eight genre groups. H1 states that at least one genre group has a significantly different mean. Test: F-test at $\alpha = 0.05$, followed by Tukey HSD post-hoc comparisons if rejected.

H3 (Logistic): H0 states that $\log(\text{Cast Popularity})$ does not change the odds of ROI exceeding 1.0. H1 states that higher cast popularity increases those odds after controlling for budget. Test: Wald test at $\alpha = 0.05$, evaluated using AUC-ROC.

Conclusion

OLS regression, ANOVA, and logistic regression have been selected and justified. Features were chosen based on EDA correlations and the Week 2 literature review. All three hypothesis sets are formally stated and ready for testing in Week 5.

References

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